SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 40%

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

NAME: K.L DINESH ESHWAR
REG NO:192425027
COURSE: COMPUTER VISION
COURSE CODE: ITA0518

28. Motion Model Evaluation

Results:

Model	**Realism**	**Speed**
A	**High**	**Low**
B	**Moderate**	**High**

Compare and evaluate based on application context.
answer this
SOLUTION: The two models can be compared based on realism and speed:

Model	Realism	Speed	Evaluation
A	High	Low	Produces more realistic motion but requires more processing time.
B	Moderat	High	Produces less realistic motion but works much faster.

Application Context:

- Model A is suitable for applications where visual quality and realism are most important, such as movies, animation, virtual reality, and game cinematics.
- Model B is better for real-time applications where speed is important, such as live games, robotics, simulations, and interactive systems.

Conclusion:
If the application prioritizes realistic motion, choose Model A. If it prioritizes fast response and real-time performance, choose Model B.

25. Model Efficiency vs Accuracy

Results:

Model	**Accuracy**	**Computation**
A	**90%**	**High**
B	**85%**	**Low**
C	**88%**	**Moderate**

Compare and evaluate for edge devices.

SOLUTION:

Model	Accuracy	Computation	Evaluation
A	90%	High	Highest accuracy but requires more processing power and energy.
B	85%	Low	Lowest accuracy but very efficient and suitable for limited hardware.
C	88%	Moderate	Good balance between accuracy and computational requirements.

For Edge Devices:

- Model A: May be unsuitable for low-power edge devices because of its high computational requirement, despite having the best accuracy.
- Model B: Best when low power consumption and fast processing are the main priorities.
- Model C: Best overall choice for many edge applications because it provides 88% accuracy with moderate computation.

Conclusion:

For edge devices, Model C is generally the best choice because it offers a good trade-off between accuracy, speed, memory usage, and power consumption. Model B should be chosen when resource efficiency is more important than accuracy.

21. Noise Impact on Detection

Results:

Noise Level	**Accuracy**
Low	**92%**
Medium	**80%**
High	**60%**

Compare performance and evaluate system robustness.

answer this

SOLUTION:
Noise impact on detection accuracy

Accuracy decreases as the noise level increases.

Noise Level	Accuracy	Performance
Low	92%	Excellent detection
Medium	80%	Moderate performance
High	60%	Poor detection

Evaluation of Robustness:

- At low noise, the system performs very well with 92% accuracy.
- At medium noise, accuracy drops to 80%, showing that noise affects detection performance.
- At high noise, accuracy falls significantly to 60%, indicating weak performance under noisy conditions.
- Therefore, the system has good performance in low-noise environments but limited robustness against high noise.

Conclusion:
The system should be improved using noise reduction, filtering, data augmentation, or noise-trained models to maintain higher accuracy in noisy environments.

1. Real-Time System Design Trade-off

A surveillance system must operate under constraints:

- **Viola-Jones: fast but less accurate (78%)**
- **YOLO: moderate speed, good accuracy (92%)**
- **Deep Learning: high accuracy (95%) but slow**

Compare the techniques and evaluate which should be selected under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

SOLUTION:

Technique	Speed	Accuracy	Advantages	Disadvantages
Viola-Jones	Very Fast	78%	Low latency and low computational cost	Lower accuracy
YOLO	Moderate	92%	Good balance between speed and accuracy	Requires more computation
Deep Learning	Slow	95%	Highest accuracy	High latency and resource requirements

Evaluation

- Viola-Jones: Best for systems with very strict latency requirements, but its 78% accuracy may not be acceptable for reliable surveillance.
- YOLO: Provides 92% accuracy with moderate speed, making it a strong choice for real-time surveillance.
- Deep Learning: Gives the highest accuracy (95%) but is too slow when immediate detection is required.

Recommended Choice: YOLO

YOLO should be selected because it provides the best trade-off between accuracy and speed. With 92% accuracy, it is considerably more reliable than Viola-Jones, while being faster and more suitable for real-time deployment than a slower deep-learning approach.

Deployment consideration:
For surveillance systems running on edge devices or systems with limited processing power, a lightweight YOLO version can be used. If the hardware is extremely limited and latency is the absolute priority, Viola-Jones may be chosen, accepting the lower accuracy.

Conclusion:
Under strict latency constraints while maintaining acceptable accuracy, YOLO is the most suitable choice because it balances real-time performance, accuracy, and computational requirements.

22. Depth Estimation Error Analysis

Results:

Method	**Error**
A	**0.5 m**
B	**1.2 m**
C	**0.8 m**

Compare and evaluate accuracy for real-world applications.

SOLUTION:

Method	Error	Evaluation
A	0.5 m	Highest accuracy because it has the lowest error
B	1.2 m	Lowest accuracy due to the highest error
C	0.8 m	Moderate accuracy

Comparison

- Method A has the lowest error (0.5 m), so it provides the most accurate depth estimation.
- Method C has a moderate error of 0.8 m, giving reasonable performance.
- Method B has the highest error (1.2 m), making it less reliable for applications requiring precise depth measurements.

Real-World Evaluation

For applications such as autonomous vehicles, robotics, navigation, and 3D reconstruction, accurate depth estimation is important for safe and reliable operation.

Conclusion:

Method A should be selected for real-world applications because it has the lowest depth estimation error (0.5 m) and therefore provides the highest accuracy and reliability. Method C can be used when moderate accuracy is acceptable, while Method B is less suitable for precision-critical applications.

24. Motion Detection under Crowd Density

Results:

Method	**Sparse Crowd**	**Dense Crowd**
Optical Flow	**High**	**Low**
Motion Models	**Moderate**	**High**

Compare and evaluate the best approach.

SOLUTION:

Method	Sparse Crowd	Dense Crowd	Evaluation
Optical Flow	High	Low	Works well when people are separated, but performance decreases in crowded scenes.
Motion Models	Moderate	High	Performs better in dense crowds and handles complex movement more effectively.

Comparison

- Optical Flow: Best for sparse crowds because individual movements can be tracked easily. In dense crowds, overlapping people make motion estimation difficult.
- Motion Models: Provide better performance in dense crowds because they can model complex and overlapping movements.

Best Approach

Motion Models are the better overall approach, especially for surveillance systems operating in crowded environments such as railway stations, stadiums, and public events.

Conclusion:

Use Optical Flow for sparse crowds when fast and simple motion detection is needed. For dense crowds, Motion Models should be preferred because they provide higher and more reliable detection performance.

6. Structure from Motion Consistency

Two datasets produce:

- **Dataset A: accurate depth but requires high-quality images**
- **Dataset B: inconsistent depth but works on low-quality images**

Compare the datasets and evaluate which setup is more practical for real-world deployment. Justify your reasoning.

SOLUTION:

Dataset	Depth Quality	Image Requirement	Evaluation
Dataset A	Accurate and consistent	Requires high-quality images	Better accuracy but sensitive to image quality
Dataset B	Inconsistent	Works with low-quality images	More flexible but less reliable

Comparison

- Dataset A provides more accurate depth estimation, which is important for applications such as 3D reconstruction, robotics, and navigation.
- However, it requires high-quality images, which may not always be available in real-world environments due to poor lighting, camera movement, blur, or low-resolution cameras.
- Dataset B can work with low-quality images, making it more flexible for real-world conditions.
- However, its inconsistent depth can lead to errors and unreliable 3D reconstruction.

Best Choice for Real-World Deployment

Dataset A is preferable when accurate depth is the main requirement and high-quality cameras are available. Its reliable depth estimation makes it more suitable for safety-critical and precision applications.

Dataset B is more practical when the system must operate with inexpensive or low-quality cameras, but additional processing or filtering may be required to improve depth consistency.

Conclusion

For most precision-oriented real-world applications, Dataset A is the better choice because accurate and consistent depth is more important than tolerance to poor image quality.

8. Model Generalization vs Overfitting

Performance:

Model	**Training**	**Testing**
A	**98%**	**70%**
B	**90%**	**88%**

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

SOLUTION:

Model	Training Accuracy	Testing Accuracy	Evaluation
A	98%	70%	High overfitting; poor generalization
B	90%	88%	Good generalization; low overfitting

Comparison

- Model A: Has very high training accuracy (98%) but testing accuracy drops to 70%. This large gap indicates overfitting. The model has learned the training data too closely and may perform poorly on unseen data.
- Model B: Has 90% training accuracy and 88% testing accuracy. The small gap shows that it generalizes well to new data.

Reliability for Deployment

Model B is more reliable for deployment because its testing performance is high and close to its training performance.

Model A has a higher training accuracy, but its 70% testing accuracy creates a significant risk of incorrect predictions when exposed to real-world data.

Conclusion: Model B should be selected because it has better generalization, lower overfitting, and more consistent performance on unseen data. For real-world deployment, testing accuracy and generalization are generally more important than achieving very high training accuracy.